HEXANY DIAMOND 5 7 11 13
A POLYRHYTHMIC, POLYTIMBRAL, MICROTONAL FIXED-MEDIA COMPOSITION

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ABSTRACT

"Hexany Diamond 5 7 11 13" is a 10-minute long fixed-media composition made with Wide Blue Sound's "Orbit" sample-playback synthesizer, living inside Native Instrument's Kontakt sampler. The piece is tuned in a scale derived from the work of Ervin Wilson, a 19 note scale called a Hexany Diamond, based on harmonics 5, 7, 11, and 13 (Narushima 2017). This paper discusses the software used for the composition, the patch, methods of performance, and the structure of the composition.

1. INTRODUCTION

"Hexany Diamond 5 7 11 13" is a 10-minute long fixed-media composition made with Wide Blue Sound's "Orbit" sample-playback synthesizer, living inside Native Instrument's Kontakt sampler. Three instances of Kontakt live inside Plogue Bidule, each with their own Step Sequencer, to provide polyrhythmic playback. This setup is performed interactively with a control panel set up in Algorithmic Arts' MusicWonk. The piece is tuned to Ervin Wilson's "Hexany Diamond 5 7 11 13" scale, a 19 note just-intonation microtonal scale, made in Marcus Hobb's "Wilsonic" app. In this patch, "Orbit" can play 4 microtonally-tuned pitches at any one time. Two passes of the patch are recorded, which, in total, play 18 pitches of the 19-note scale. The piece features polyrhythmic alternations of Orbits 4 sample players, building up to an 18 note cluster near the end.

2. ORBIT - THE INSTRUMENT

Orbit is a sample-playback instrument from Wide Blue Sound of Los Angeles. It is one of a number of recent sample-playback systems that feature more complex pre-made samples in an effort to assemble more complex timbres than simple analogue synthesis or acoustic-instrument playback. Other examples of this kind of synthesizer are "Infundibulum" from Sound Dust, "Kepler Orchestra" from Spitfire Audio, and "Cataract" from Glitchmachines. Living inside Kontakt, Orbit has four sample players. These four can play back from a library of over 100 fairly long samples. Each of these samples features some kind of sonic evolution. Each sample player can be tuned precisely, filtered, panned, etc. Any selection of the four players can be alternated between with three different transitions – different



Figure 1. Wide Blue Sound: Orbit Sample Player

envelopes used to cross-fade between the players. The rate of these transitions is also controllable, either with the on-screen control, or with an external MIDI modulator. A "depth" control will allow the players to fade from alternating envelopes to constant playing. Figure 1 shows the user interface of Orbit.

The sounds made by Orbit all feature a kind of sonic evolution. Even the basic sine-wave-like timbre, called "Pure," has a changing spectrum over time. Other sounds such as "Shimpering" have more pronounced harmonics, but less long-term sonic evolution. Each of the over 100 sounds available in Orbit has some kind of sonic evolution and each is at least 20 seconds long. Most of the sounds, despite their evolution, still have a sense of being centred on a pitch, with focus on a more or less harmonic spectrum. Wide Blue Sound has another unit, with the same interface, but with a different set of samples, many of which are more "texturally" oriented rather than pitch-centric. This synth is called "Eclipse." With four sample players, one can alternate between up

to 4 different timbres. The kind of envelope used for the transitions, from sharp (sawtooth or square wave envelopes) to smooth (sine wave envelopes). The speed of alternation can also be changed in real time. Further, the depth of this alternating envelope can also be changed. With "depth" set to 100, the four waveforms are heard as distinct sounds. With depth set to 0, then all four waveforms are heard as a sustained poly-timbral chord. Further, the pitch of each sample player can be microtonally detuned by cents to a tolerance of 2 decimal points. So a typical sound on Orbit can involve alternating between four different evolving, differently tuned samples. The speed of alternating of the timbres can be changed, as can the depth of the alternation, up to all four timbres being heard together in a sustained polytimbral microtonal chord.

3. THE PATCH FOR EACH OF TWO PASSES

The piece was made in two real-time passes, each using the same patch, but with different timbres and tunings for each pass. Two patches were used and they are shown in Figure 2. The real-time performance interface was made with Algorithmic Arts "MusicWonk," and the overall patch was made with Plogue's "Bidule." The MusicWonk controller patch (top) has an on/off switch for everything, then three sliders, one each for "Depth" on each instance of Orbit, then an alternation envelope setting for each of the three Kontakt instruments, then three speed selectors, one for each of the three Orbit's alternation speeds.

The Plogue patch (bottom) has a MIDI in (from MusicWonk), three Step Sequencers (one for each of the three Kontakts, the three Kontakts, and a mixer and output. Each instance of Kontakt has a different version of Orbit, each micro-tuned to a different microtonal chord, and has a different selection of timbres. The three step sequencers continually change tempo between 20 and 240 bpm independently, producing polyrhythms. These are controlled by three sine-LFOs in MusicWonk.

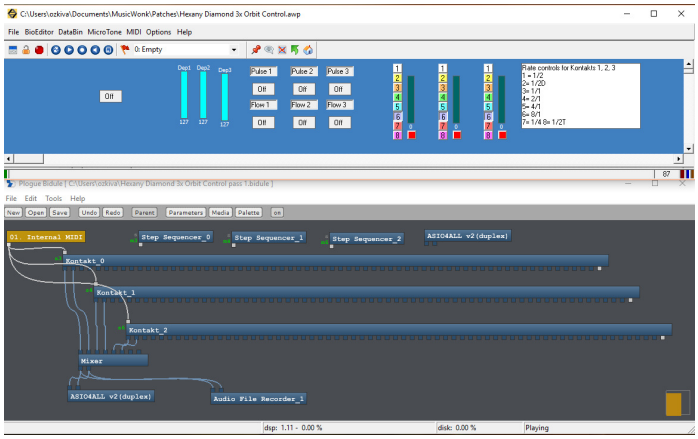


Figure 2. (top) Real-time controller interface made with MusicWonk. (bottom) Overall patch for each pass of the piece made with Plogue Bidule.

3.1 Controlling the Speed of the Alternating Timbres

In Figure 3, in the top of the Kontakt player, you see a window called Sync. In this, Step Sequencer 1 is selected. At the top left, you see Step Sequencer 1 open. It continually varies between 20 and 240 bpm. This means the alternating of the timbres between the four sample players in Orbit is continually accelerating and

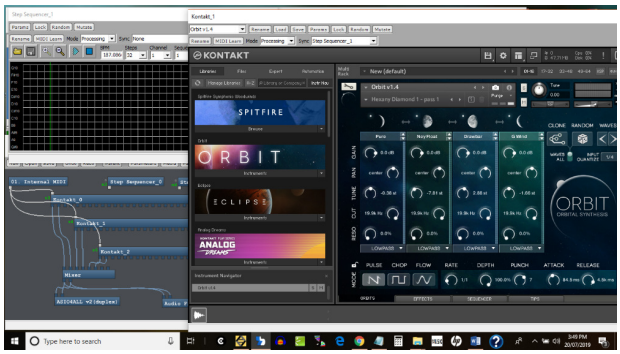


Figure 3. The connection of the MIDI step sequencers to the Kontakt Player for real time control of alternation speed between the separate timbres.

decelerating. With three of these going independently, quite complex polyrhythms are generated. In fact, with three generators using different tempi changing between 20 and 240 bpm, and the rate of acceleration and deceleration of the three tempi all proceeding at different rates, the polyrhythms are beyond my capability to predict, or describe, and this is a kind of irrationally relating rhythms that I have been assembling, and learning how to listen to since at least the early 1970s. With the two passes of the patch mixed together, 6 different tempi of alternating timbres are heard simultaneously. Occasionally, the Depth controls on the Orbits are moved to 0. This makes a sustained chord of the 4 pitches in any instance of Orbit. At about 8 minutes in, all the Depth controls are moved to 0, making a very thick microtonal chord.

4. THE TUNING

This piece uses a scale based on the work of Ervin Wilson.



Figure 4. Ervin Wilson, 2010. Portrait by Catherine Schieve.

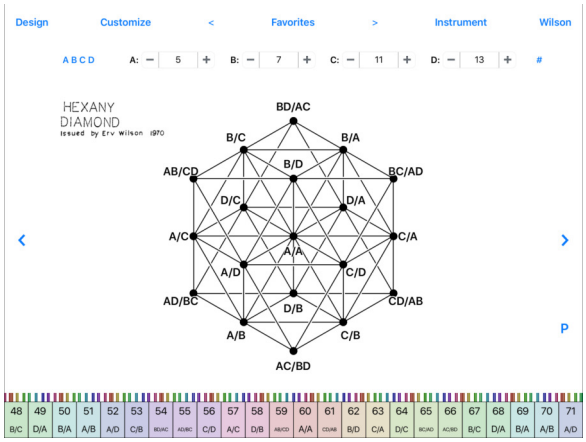


Figure 5. Hexany Diamond page from Marcus Hobbs' Wilsonic iOS app.

Ervin Wilson (1928-2016) was a music theorist in Los Angeles. Student of Harry Partch, he extended just intonation (and other) tuning ideas. The Hexany is a 6 element Combination Product Set featuring tuning ratios made from all combinations of 2 elements out of 4. Usually these are numbers of harmonics. A Hexany Diamond is made of the initial Hexany, plus its inversion, plus all 6 possible ratios made from all the 2 element products of the set. Plus a 1/1 central tone from which all the other pitches derive. This makes a 19 note scale. Figure 5 shows the Hexany in Marcus Hobbs free Wilsonic iOS app.

4.1 The Hexany Diamond used in this Piece.

Since a Hexany Diamond is made of operations on 4 factors, usually harmonics, absolutely arbitrarily, I took 4 prime-numbered harmonics, 5, 7, 11, and 13. I used prime numbers to make sure that there were no harmonic duplicates in the scale. Apart from that, they could be any set of four prime-numbered harmonics. In other pieces of this set "Hexany Diamonds and Others (2019)", I used other sets of prime-numbered harmonics to make different scales for those pieces. I take my cue from Stefan Wolpe, with his essay "Any Bunch of Notes," in which he maintains that any arbitrary collection of pitches can be used to make a composition – one then finds the relationships resulting from one's arbitrary decision and works with those (Wolpe, 1982). Using the chosen four factors, make a Hexany out of these, using all the ratios of any two elements out of the 4 given. The following lines show how the ratios are derived from each set of two elements from the four given harmonics. The final line shows the 6 pitches of the hexany in cents.

A/B = 5/7 = 10/7 = 617c;

A/C=5/11=20/11=1035c;

A/D=5/13=20/13=746c;

B/C=7/11=14/11=418c;

B/D=7/13=14/13=128c;

C/D=11/13=22/13=911c.

Hexany = 128c 418c 617c 746c 911c 1035c

These next lines show how the inversion of the hexany is derived. Notice that above, the first ratio is A/B. To get the inversion, simply reverse the terms: B/A.

Inversion

B/A=7/5=583c;

C/A=11/5=11/10=165c – etc.

Hexany Inversion = 165c 289c 454c 583c 782c 1072c

Finally, the 6 possible ratios of all combinations of the four chosen harmonics are calculated.

Combinations: AB/CD; AC/BD; AD/BC; BD/AC; BC/AD; CD/AB

As an example C*D/A*B = 11*13/5*7 = 143/35 = 143/140 = 37c

Combination Hexany: 37c 293c 328c 872c 906c 1163c

Refer to Hexany Diamond diagram in Figure 5 for all relationships possible within this scale.

4.2 The Complete Scale listed in Scala format.

0:	A/A	1/1	0.000000	unison
1:	CD/AB	143/140	36.705984	
2:	B/D	14/13	128.298245	2/3-tone
3:	C/A	11/10	165.004228	4/5-tone, Ptolemy's second
4:	D/C	13/11	289.209719	tridecimal minor third
5:	BC/AD	77/65	293.302473	
6:	AC/BD	110/91	328.278088	
7:	B/C	14/11	417.507964	undecimal diminished fourth or major third
8:	D/A	13/10	454.213948	tridecimal semi-diminished fourth
9:	B/A	7/5	582.512193	septimal or Huygens' tritone, BP fourth
10:	A/B	10/7	617.487807	Euler's tritone
11:	A/D	20/13	745.786052	tridecimal semi-augmented fifth
12:	C/B	11/7	782.492036	undecimal augmented fifth
13:	BD/AC	91/55	871.721912	
14:	AD/BC	130/77	906.697527	
15:	C/D	22/13	910.790281	tridecimal major sixth
16:	A/C	20/11	1034.995772	large minor seventh
17:	D/B	13/7	1071.701755	16/3-tone
18:	AB/CD	280/143	1163.294016	
19:	A/A	2/1	1200.000000	octave

4.3 How the Scale is used in this piece.

Here's how the instances of Orbit are tuned – (or as closely as possible that the fine tuning control of Orbit will allow) – remember, there are three instances of Orbit being used in each pass of this piece, and each instance of Orbit allows only 4 precisely tuned tones at a time, which means that each pass of the piece will have 3 tetrachords, for a total of 12 pitches (out of 19) being used in each pass. Octave placement is determined by if the ratio in the Hexany Diamond is, say 7/11 (lower octave) or 11/7 (normal octave) or an octave below or above, depending on the ratio:

Pass 1 Instance 1 BD/AC 871c; B/A 582c; B/D 128c; B/C 417c – chord 1:

Pass 1 Instance 2 AB/CD -36c; B/C -782c; D/C 289c; A/C -1365c – chord 2:

Pass 1 Instance 3 BC/AD 293c; B/A 582c; D/A 1654c; C/A 1365c

Pass 2 Instance 1 AD/BC -293c; A/C -1365c; A/D -1654c; A/B -582c

Pass 2 Instance 2 AC/BD -871c; CD 782c; D/B 1071c; A/B -582c

Pass 3 Instance 3 CD/AB 1236c; C/A 1365c; C/D -289c; C/B 782c

This makes a total of 6 tetrachords for the entire piece. Each pass uses 12 pitches, which when combined encompass 18pitches from the 19-tone scale. Only the 1/1 A/A is left out.

Again, two passes were made in real-time and mixed together. With the MusicWonk patch, changes were improvised in depth, individual instance rhythm and saw/sine voice alternation envelopes, using the patch shown in the upper half of Figure 2. As mentioned earlier, by 8 minutes in all depths of the alternating timbres were set to 0 so that the full cluster chord was heard. At 9 minutes, all depths on all channels were faded back up to 100 so that there would be a return to the polyrhythmic texture which started things off for the final minute of the piece. My normal compositional practice would usually involve setting up some kind of algorithmic process to control the details of the piece as they unfolded. With Orbit, and the given tuning, I felt that the patch itself embodied a kind of algorithmic process, and that all I needed to do for each pass was to improvise a selection of alternations between polyrhythmic textures (the irrational polyrhythms themselves produced by low frequency oscillators controlling the clocks on the Plogue sequencers), and full textured chords in order to create a structurally satisfying structure. The 10 minute piece which resulted from this is part of a much longer piece "Hexany Diamonds and Others" which was produced in 2019, using, among other resources, multi-timbral instruments referred to in this text (Infundibulum, Kepler Orchestra, Orbit, Eclipse) and a number of other Hexany Diamond scales among other tunings. The entire piece lasts 130 minutes and consists of 13 movements.

5. REFERENCES

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Wolpe, Stefan and Clarkson, Austin *"Any Bunch of Notes": A Lecture*, Perspectives of New Music, Vol. 21, No. 1/2 (Autumn, 1982 - Summer, 1983), pp. 295-310

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Infundibulum http://dulcitone1884.virb.com/infundibulum2 (accessed 7 Sept 2019)

Kepler Orchestra https://www.spitfireaudio.com/shop/a-z/kepler-orchestra/ (accessed 7 Sept 2019)

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